FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2019

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. If A × B = C, then which of the following statements is wrong -
(A) C ⊥ A (B) C ⊥ B
(C) C ⊥ (A + B) (D) C ⊥ (A × B)

2. The dimensional formula of a physical quantity x is $\left[M^{-1}L^{3}T^{-2}\right]$. The error in measuring the quantities M, L, and T are 2%, 3% and 4%. The maximum percentage of error that occurs in measuring, the quantity x is:
(A) 9 (B) 10
(C) 14 (D) 19

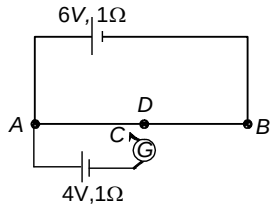
3. A rocket is fired vertically up from the ground with a resultant vertical acceleration of $10\,m/s^2$. The fuel is finished in 1 minute and it continues to move up.
(A) The maximum height reached.
(B) After how much time from then will the maximum height be reached (Take g = 10 m/s^2)
(A) 36km, 1 min (B) 6km, 1 min
(C) 36km, 1 sec (D) 36 km, 1 sec

4. An object of 6 kg is in the state of limiting equilibrium on an inclined plane having angle of inclination $30°$. Now if the angle of inclination is increased to $60°$, then the extra force along the plane to be applied on the object, to keep it in equilibrium will be (take g = 10 m/s^2)
(A) 3 kg wt (B) $2\sqrt{3}$ kg wt
(C) $\sqrt{3}$ kg wt (D) $3\sqrt{3}$ kg wt

5. A particle moving in a straight line is acted by a force, which works at a constant power and changes its velocity from u to v in passing over a distance x. The time taken will be:
(A) $x\left(\frac{v-u}{v^2+u^2}\right)$ (B) $x\left(\frac{v+u}{v^2+u^2}\right)$
(C) $\frac{3}{2}\,x\left(\frac{v^2-u^2}{v^3-u^3}\right)$ (D) $x\left(\frac{v}{u}\right)$

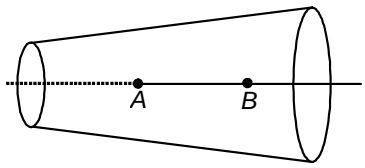
Space for Rough work

6. A 6 volt battery of internal resistance 1Ω is connected across a uniform wire AB of length 100 cm. The positive terminal of another battery of emf 4V and internal resistance 1Ω is joined to the point A as shown. Find the distance of point D from A if there is no current through galvanometer. (resistance of $AB = 5\Omega$)



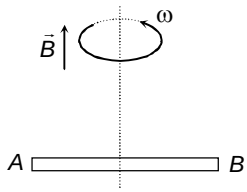
(A) 80 cm (B) 60 cm
(C) 75 cm (D) 40 cm

7. A resistance of frustum shape is shown in figure. If a current *i* passes through the resistance, the electric field at *A* and *B* are related as



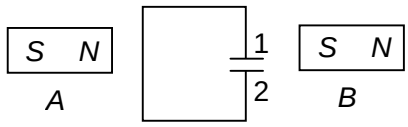
(A) $E_A > E_B$
(B) $E_B > E_A$
(C) $E_A = E_B$
(D) there is no relation

8. A conducting rod of length 2*l* is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is



(A) $B\omega l^2$ (B) $\frac{1}{2}B\omega l^2$
(C) $\frac{1}{8}B\omega l^2$ (D) zero

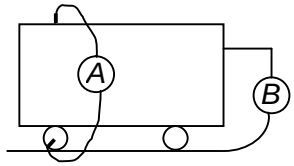
9. Two bar magnet moving with same speed in the given figure. Choose correct statement.



(A) Plate (1) will be +ve relative to (2) if *A* moving *B* moving towards left.
(B) Plate (1) will be +ve relative to (2) if *A* moving towards left and *B* moving towards right.
(C) Charge on capacitor increase if *A* and *B* are at rest.
(D) None of these

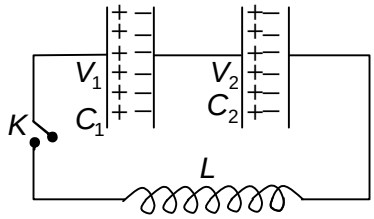
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10. A flat, rail road car with a wooden base moves left with some velocity. A millivoltmeter *A* is connected across the axle of the car and is in the car. Another voltmeter *B* is connected across the rails of the car. Then



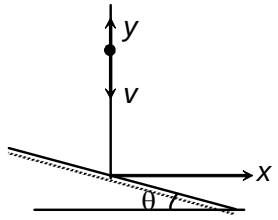
(A) Reading of *A* and *B* = zero
(B) Reading of *A* = zero and *B* is non zero.
(C) Reading of *B* = zero and *A* and is no zero.
(D) Reading of *A* and *B* are non zero.

11. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance *L* as shown in the figure. Initially key *k* is open. Now key *k* is closed and current in the circuit starts increasing. When current in the circuit is maximum



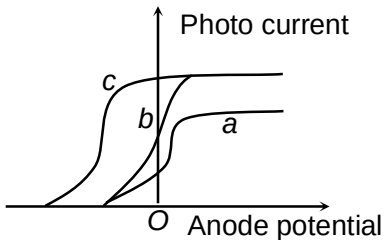
(A) charge on both the capacitors is same
(B) induced emf in the inductor is zero
(C) potential difference across both the capacitors is half of the induced emf
(D) electrostatic potential energy stored in both the capacitors is same

12. An object is moving towards a mirror with a velocity *v* as shown in figure. If the collision between the mirror and the object is perfectly elastic, then the velocity of the image after collision with mirror in vector form is



(A) $-v\hat{j}$
(B) $-v\cos 2\theta\hat{j} + v\sin 2\theta\hat{i}$
(C) $-v\hat{i}$
(D) $-v\cos 2\theta\,\hat{j} - v\sin 2\theta\hat{i}$

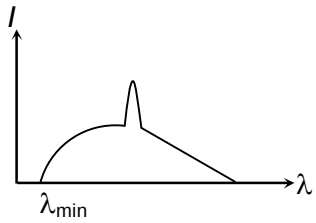
13. The figure shows the variation of photo current with anode potential for a photo-sensitive surface for three different radiations. Let I_a, I_b and I_c be the intensities and f_a, f_b and f_c be the frequencies for the curves *a*, *b* and *c* respectively;



(A) $f_a = f_b$ and $I_a \neq I_b$
(B) $f_a = f_c$ and $I_a = I_C$
(C) $f_a = f_b$ and $I_a = I_b$
(D) $f_b = f_c$ and $I_b = I_c$

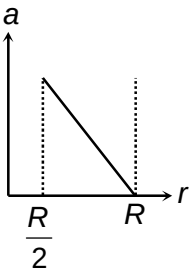
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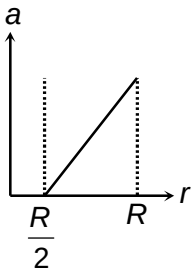
14. An X-ray tube has three main controls.
(i) the target material (its atomic number Z)
(ii) the filament current (I_f) and
(iii) the accelerating voltage (V)
Figure shows a typical intensity distribution against wavelength.

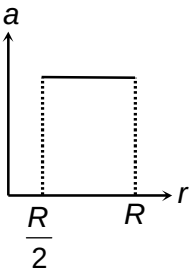


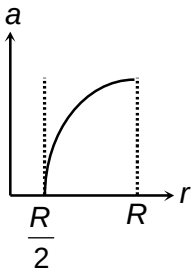
Which of the following is incorrect?
(A) The limit λ_{min} is proportional to V^{-1}
(B) The sharp peak shifts to the right as Z is increased
(C) The penetrating power of X ray increases if V is increased
(D) The intensity everywhere increases if filament current I_f is increased

15. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is

(A)


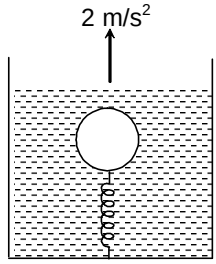
(B)


(C)


(D)


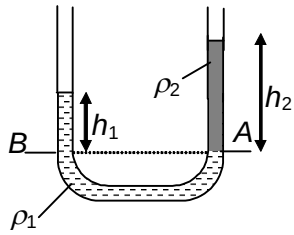
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16. A ball of mass 10 kg and density 1 gm/cm^3 is attached to the base of a container having a liquid of density 1.1 gm/cm^3, with the help of a spring as shown in the figure. The container is going up with an acceleration 2 m/s^2. If the spring constant of the spring is 200 N/m, the elongation in the spring is



(A) 2 cm (B) 4 cm
(C) 6 cm (D) 8 cm

17. Two immiscible liquids are poured in a *U*-tube having densities ρ_1 and ρ_2. The ratio of height of the liquids above their interface $\left(h_1 / h_2\right)$ is



(A) directly proportional to their densities
(B) inversely proportional to their densities
(C) directly proportional to square of their densities
(D) equal

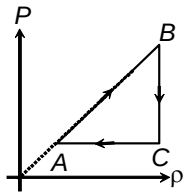
18. Two particles are performing simple harmonic motion with same amplitude and same frequency. When they are at same distance from mean position on opposite sides, their speeds are one fourth of maximum speed. Also at these positions their direction of velocities are opposite, the phase difference between the two simple harmonic motions is

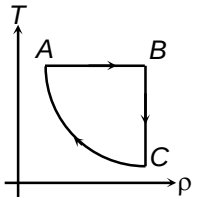


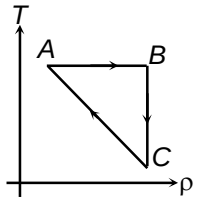
(A) $\frac{\pi}{2}$ (B) π
(C) $\frac{3\pi}{2}$ (D) $\frac{5\pi}{4}$

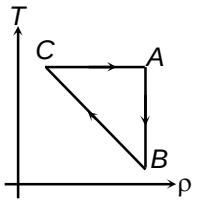
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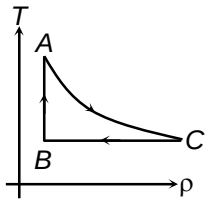
19. Pressure versus density of an ideal gas is shown in the graph. Then the correct temperature versus density of gas graph is



(A) 

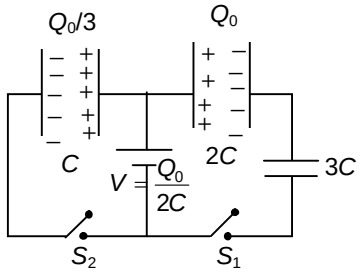
(B) 

(C) 

(D) 

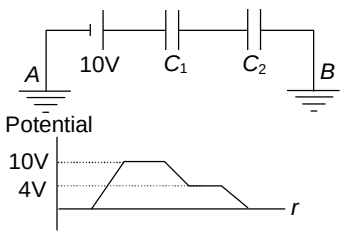
20. P-V diagram of a diatomic gas is straight line parallel to P-axis. The molar heat capacity of the gas in the process will be
(A) 4*R* (B) 2.5 *R*
(C) 3*R* (D) 4*R*/3

21. In the given circuit, the initial charges on the capacitors are shown in the figure. The charge flown through the switches S_1 and S_2 respectively after closing the switches are



(A) zero, $\frac{Q_0}{6}$ (B) $\frac{Q_0}{5}$, $\frac{Q_0}{2}$
(C) zero, $\frac{Q_0}{2}$ (D) $\frac{3}{5}Q_0$, $\frac{Q_0}{6}$

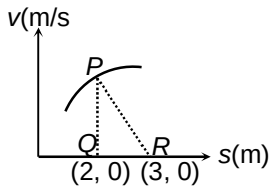
22. Figures shows two capacitors C_1 and C_2 connected with 10 V battery and terminal *A* and *B* are earthed. The graph shows the variation of potential as one moves from left to right. Then the ratio of C_1/C_2 is



(A) 5/2 (B) 2/3
(C) 2/5 (D) 4/3

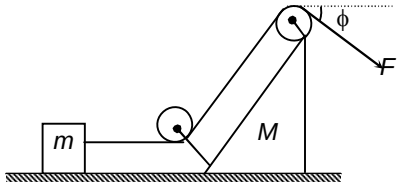
Space for Rough work

23. Velocity versus displacement curve of a particle moving in straight line is shown in the figure. From a point *P*, a line is drawn perpendicular to displacement axis and line *PR* is drawn normal to the curve at *P*. The instantaneous acceleration of the particle at point *P* is



(A) 1 m/s^2 (B) 2 m/s^2
(C) 3 m/s^2 (D) 2.5 m/s^2

24. In the given arrangement pulleys and string are massless and frictionless and all surfaces are smooth. Find the magnitude of acceleration of wedge of mass *M*.



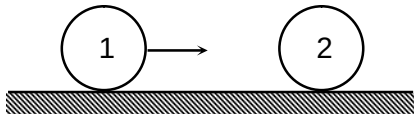
(A) $\frac{F\cos\phi}{M}$

(B) $\frac{F(1-\cos\phi)}{M}$

(C) $\frac{F}{m+M}$

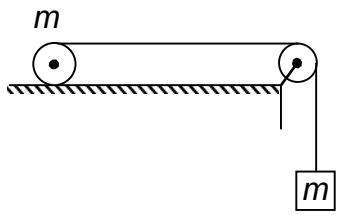
(D) $\frac{F\cos\phi}{m+M}$

25. Ball 1 collides with an another identical ball 2 at rest as shown in figure. For what value of coefficient of restitution *e*, the velocity of second ball becomes two times that of 1 after collision?



(A) 1/3 (B) 1/2
(C) 1/4 (D) 1/6

26. In the given figure a ring of mass *m* is kept on a horizontal surface while a body of equal mass '*m*' attached through a string, which is wounded on the ring. When the system is released the ring rolls without slipping. Consider the following statements and choose the correct option.



(i) acceleration of the centre of mass of ring is $2g/3$

(ii) acceleration of the hanging particle is $4g/3$

(iii) frictional force (on the ring) acts along forward direction

(iv) frictional force (on the ring) acts along backward direction

(A) statement (i) and (ii) only (B) statement (ii) and (iii) only
(C) statement (iii) and (iv) only (D) none of these

Space for Rough work

27 A brass scale is graduated at 10ºC. What is the true length of a zinc rod which measures 60.00 cm on this scale at 30ºC. Coefficient of linear expansion of brass = $18 \times 10^{-6}\ K^{-1}$.

(A) 60.02 cm (B) 50.02 cm
(C) 65.04 cm (D) 55.05 cm

28. A uniform rope of length 12 metre & mass 6 kg, is swinging vertically from rigid base. From its free end, one 2 kg mass is attached. At its bottom end one transverse wave is produced of wavelength 0.06 metre. At upper end of rope, wavelength will be-

(A) 1.2 m (B) 0.12 m
(C) 0.12 cm (D) 0.12 mm

29. A very small sphere of mass 80 g having a charge q is held at a height 9 m vertically above the centre of a fixed non conducting sphere of radius 1 m, carrying an equal charge q. When released it falls until it is repelled just before it comes in contact with the sphere. Calculate the charge q.

[$g = 9.8\ m/s^2$]

(A) $28\mu C$ (B) $26\mu C$
(C) $27\mu C$ (D) $25\mu C$

30 In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is 3×10^{-5} cm. If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.

(A) 7000Å (B) 5000Å
(C) 6000Å (D) 4000Å

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following statement is **INCORRECT** about the homogeneous reaction? $aA + bB \rightarrow cC + dD$.

(A) The relation $-\frac{1}{a}\frac{dn_A}{dt} = \frac{1}{c}\frac{dn_C}{dt}$ need not hold if the reaction has more than one step and concentration of reaction intermediates are not small.

(B) The rate of reaction $= k[A]^a[B]^b$ if the above reaction is elementary reaction.

(C) The rate of reaction = $-\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$.

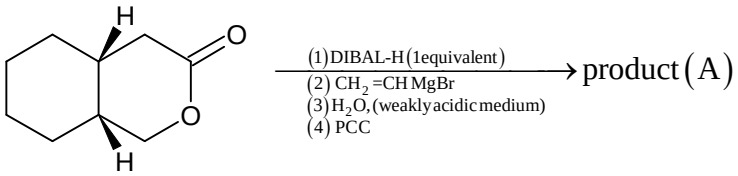
(D) If concentration of A reduces to half in time x seconds than concentration of B reduces to half in time $\frac{b}{a}x$ seconds regardless of initial concentrations of A & B and order of reaction w.r.t. A & B

32. Which species is linear at the central atom?

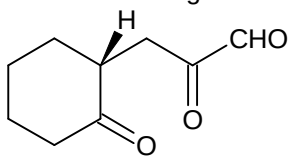
(A) CO_3^{2-} (B) $[S_3\,Sn\,S\,Sn\,S_3]^{6-}$

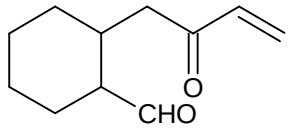
(C) $Me_3SiOSiMe_3$ (D) SC_3S

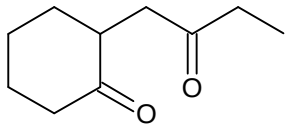
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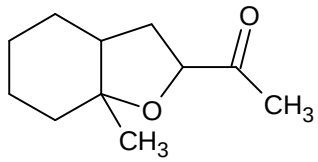
33. 

Which of the following is correct structure of 'A' ?

(A) 

(B) 

(C) 

(D) 

34. Which of the following statement is correct about various periodic properties?
(A) The electron affinity of an atom under consideration is equal to the negative of the electron gain enthalpy at any temperature.
(B) The electronegativity of any given element is not constant; it varies depending on the element to which it is bound.
(C) The first ionization enthalpy values of group 13 elements varies as $B > Al > Ga > In > Tl$.
(D) An element with ground state electronic configuration $[Xe]4f^{14}5d^{1}6s^{2}$ belongs to d-block in the modern periodic table.

35. Which of the following statement is correct about various properties?
(A) At 1 atm pressure boiling temperature is called standard boiling point while at 1 bar pressure boiling point is called normal boiling point.
(B) Standard boiling point of the liquid is slightly greater than the normal boiling point.
(C) Boiling does not occur when liquid is heated in a closed vessel.
(D) Vapour pressure of various compounds at 300K is in order, diethyl ether > $CCl_4 > H_2O >$ ethyl alchol

Space for Rough work

36. Calculate bond enthalpy (in kJ/mol) of C-Cl in $CCl_4(g)$ from the following

Data:

$\Delta_{vap.}H^\circ_{CCl_4} = 30.5\,kJ/mol \quad \Delta_f H^\circ_{(CCl_4,l)} = -135.5\,kJ/mole$

$\Delta_a H^\circ(C) = 715\,kJ/mol \ (i.e.\ enthalpy\ of\ atomisation)$

$\Delta_a H^\circ_{(Cl_2)} = 242\,kJ/mole$

(A) 326 (B) 333.625

(C) 356.5 (D) 447

37. For a system (such as an atom or a molecule whose energy does not change with time) the Schrodinger equation is written as $\hat{H}\psi = E\psi$. Which of the following is **INCORRECT** about the Schrodinger equation?

(A) $\hat{H}$ is a mathematical operator called Hamiltonian.

(B) The solution of this equation gives E and ψ.

(C) Schrodinger gave a recipe of constructing $\hat{H}$ operator from the expression for the total energy of the system.

(D) The total energy of the system takes into account the kinetic energies of all the sub-atomic particles, attractive potential between the electrons and nuclei but excludes repulsive potential among the electrons and nuclei individually.

38. Aqueous solution of boric acid is treated with salicylic acid. Which of the following statement is correct for the product formed in the above reaction?

(A) No product will be formed because both the acid

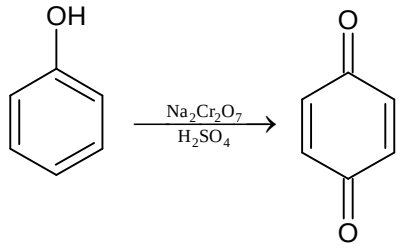
(B) Product is 4-coordinated complex and optically resolvable

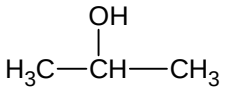
(C) Product is 6-coordinated complex and optically non-resolvable

(D) There are two ring only which are five membered

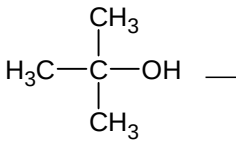
Space for Rough work

39. Which of the following reaction **INCORRECTLY** matched?

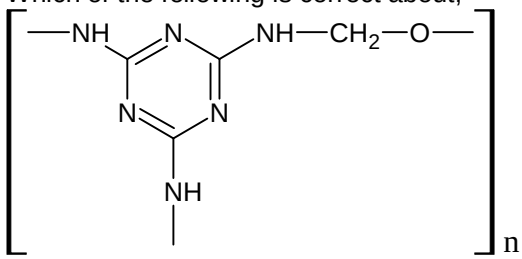
(A)  $\left(\text{oxidation}\right)$

(B)  $\left(\text{oxidation}\right)$

(C) $CH_3-CH=CH-CH_2-OH \xrightarrow{PCC} CH_3-CH=CH-CHO$ $\left(\text{dehydrogenation}\right)$

(D)  $\left(\text{oxidation}\right)$

40. Which of the following is correct about,

 polymer?

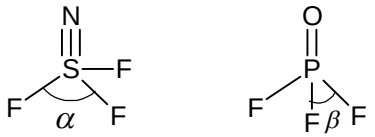
(A) Polymer is formed by addition polymerisation of melamine and formaldehyde.
(B) It is used in manufacturing of unbreakable crockery.
(C) It is a thermoplastic polymer i.e. capable of repeatedly softening on heating and hardening on cooling.
(D) It is a linear polymer.

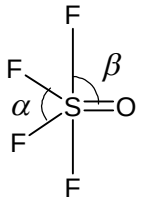
Space for Rough work

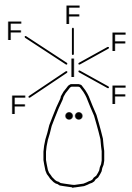
41. Which of the following statement is correct about antimicrobials?
(A) Ampicillin and amoxycillin are narrow spectrum antibiotics.
(B) Antibiotics have both cidal (killing) effect and static (inhibitory) effect on microbes.
(C) A substance cannot act as an antiseptic as well as disinfectant by varying the concentration.
(D) A mixture of chloroxylenol and terpineol, tincture of iodine, iodoform, boric acid etc are used as antiseptics.

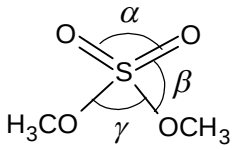
42. Which of the following Vitamin is not matched with its correct scientific name?
(A) Vitamin B_1=Threonin (B) Vitamin B_2=Riboflavin
(C) Vitamin B_6=Pyridoxine (D) Vitamin C=Ascorbic acid

43. Which of the following properties about bond angle and bond lengths is correct for the given compound?

(A)  $\alpha > \beta$

(B)  $\alpha = 120°$ *and* $\beta > 90°$

(C)  all I-F bonds are of equal length because Iodine is sp^3d hybridized.

(D)  $\alpha > \beta > \gamma$

Space for Rough work

44. Calculate the molar solubility of $Ni(OH)_2$ in 0.1 M $NaOH$. The solubility product of $Ni(OH)_2$ is 2×10^{-15}. (Given: $\sqrt[3]{0.5}=0.79$)

(A) $2\times10^{-13}M$ (B) $2\times10^{-14}M$

(C) $7.9\times10^{-6}M$ (D) $7.9\times10^{-4}M$

45. Calculate ΔG for the following reaction

$H_2O(l) \longrightarrow H_2O(g)$

$(2atm, 373K) \quad (2\,atm, 373K)$

(Given: $0.0821\times373\times\ln 2 = 21.23$ atm-ltr, molar volume of water at 2 atm , 373 K=18ml)

(A) $\Delta G = 0$ (B) $\Delta G > 0$

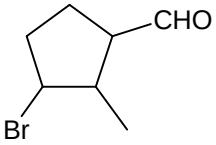
(C) $\Delta G < 0$ (D) None of these

46. Which of the following is not a magnesium ore?

(A) Carnalite (B) Dolomite

(C) Magnesite (D) Malachite

47. Which of the following is the correct IUPAC name of the compound?



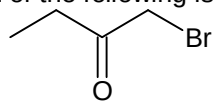
(A) 2-methyl-3-bromocyclopentane carbaldehyde
(B) 1-formyl-2-methyl-3-bromocyclopentane
(C) 3-bromo-2-methyl cyclopentane carbaldehyde
(D) 1-bromo-2-methyl cyclopentane-3-carbaldehyde

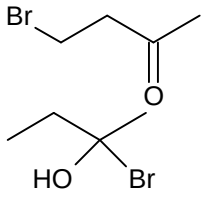
48. Which of the following is correct about defects in solid crystals?

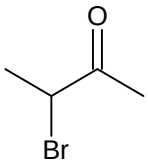
(A) Vacancy defects and interstitial defects are two main types of non-stoichiometric defects.
(B) If molten NaCl containing a little amount of MCl_3(ionic) is crystallised, the cationic vacancies thus produced are equal to twice the number of M^{3+} ions.
(C) Zinc oxide is white in colour at room temperature, on heating it turns green and its formula becomes $Zn_{1-x}O$
(D) Frenkel defect is shown by ionic substance in which the cation and anion are of almost similar sizes.

Space for Rough work

49. Which of the following is the major product in the acid catalysed bromination of butan-2-one?

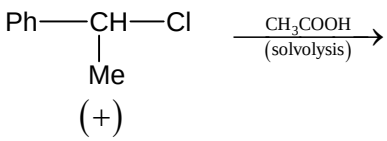
(A) 

(B) 

(C) 

(D)

50. Adding the much more powerful nucleophile CH_3COO^- (as CH_3COONa) to the reactant mixture was found to result in:



88% racernisation

12% net inversion

(A) No increase in the overall reaction rate
(B) Increase in the proportion of net inversion.
(C) The inversion that is observed does stem from part of the overall reaction proceeding via an SN^2 path
(D) A mixed $S_{N^1}+S_{N^2}$ pathway is operating in the above reaction.

51. Which of the following is correct about conductivity?
(A) $0.1M\ HCl > 0.01M\ CH_3COOH > pure\ water > 0.1M\ CH_3COOH$ (Conductivity order of aqueous solution)
(B) $Copper > Glass > Si > Fe$ (Conductivity order)
(C) The conductivity of ionic solution depends on size of ions produced, concentration of electrolyte, nature of solvent & its viscosity, nature of electrolyte.
(D) Molar conductivity of weak electrolytic solution increase on increasing the concentration of electrolyte.

Space for Rough work

52. When colloidal solution is viewed under a powerful ultra microscope, the colloidal particles appear to be in a state of continuous zigzag motion. Which of the following is correct about the zigzag motion i.e. Brownian motion?
(A) It depends on the nature of the colloid
(B) It depends on the size of the particles and viscosity of the solution
(C) It is independent of the mass of the particles.
(D) Lesser the viscosity slower in the Brownian motion

53. The correct sequence of CO bond length in the given compound is:
(I) $Fe(CO)_5$ (II) CO
(III) $H_3B \leftarrow CO$ (IV) $\left[Mn(CO)_5\right]^-$
(A) II>IV>III>I (B) II>III>I>IV
(C) III>I>IV>II (D) IV>I>II>III

54. Which of the following complex is optically active?
(A) trans-$\left[Co(NH_3)_4Cl_2\right]^+$ (B) Cis-$\left[Co(NH_3)_2(en)_2\right]^{3+}$
(C) trans-$\left[Co(NH_3)_2(en)_2\right]^{3+}$ (D) Cis-$\left[CoCl_3(NH_3)_3\right]$

55. Which of the following salt will not give positive brown ring test?
(A) $Cu(NO_3)_2$ (B) $Pb(NO_3)_2$
(C) $Zn(NO_3)_2$ (D) $Mg(NO_3)_2$

56. Which of the following is not a correct transformation?
(A) $2PCl_5(s) \longrightarrow \left[PCl_4\right]^+\left[PCl_6\right]^-$
(B) $\left[Al(H_2O)_6\right]^{3+} + H_2O \longrightarrow \left[Al(H_2O)_5(OH)\right]^{2+} + H_3O^+$
(C) $BiF_3 + H_2O \longrightarrow H_3BiO_3 + 3HF$
(D) $XeF_6 + NaF \longrightarrow Na\left[XeF_7\right]$

57. PH_3 can be obtained by:
(A) Heating phosphorous acid
(B) Reacting white P_4 with hot concentration NaOH
(C) Hydrolysis of calcium phosphide
(D) All of the above

Space for Rough work

58. Prussian blue is a deep blue pigment containing Fe^{2+}, Fe^{3+} and CN^- ions. It has the formula $Fe_7(CN)_{18}$. How many Fe^{2+} and Fe^{3+} ions are there per formula unit?

(A) $3Fe^{2+}$ and $4Fe^{3+}$ (B) $4Fe^{2+}$ and $3Fe^{3+}$

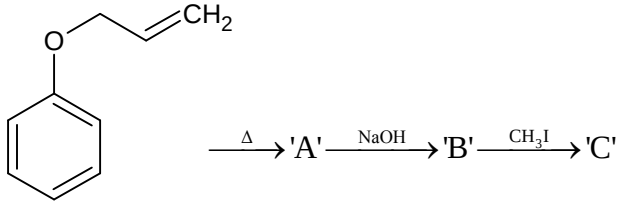
(C) $5Fe^{2+}$ and $2Fe^{3+}$ (D) $6Fe^{2+}$ and $1Fe^{3+}$

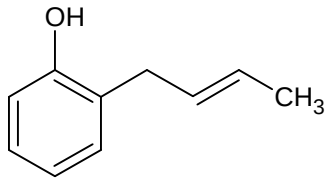
59. A small piece of Na metal is placed in an ignition tube and small quantity of the compound is added and tube is heated strongly. The red hot tube is plunged into 10 ml distilled water contained in a mortar is crushed and solution is filtered. The filtrate is called sodium extract. To small quantity of extract few crystals of Mohr's salt are added and boiled on adding dilute H_2SO_4 to it, an intense blue is observed. The element present in the compound is?

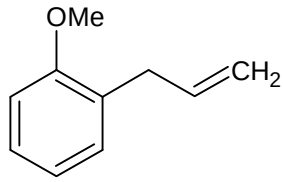
(A) S (B) halogen

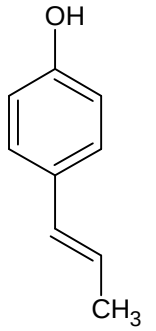
(C) nitrogen (D) phosphorus

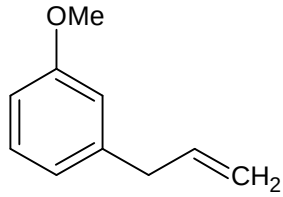
60. The product 'C' in the given reaction sequence is:



(A) 

(B) 

(C) 

(D) 

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Which of the following function is surjective but not injective?

(A) $f : R \to R,\ f(x) = x^4 + 2x^3 - x^2 + 1$

(B) $f : R \to R,\ f(x) = x^3 + x + 1$

(C) $f : R \to R^+,\ f(x) = \sqrt{1 + x^2}$

(D) $f : R \to R,\ f(x) = x^3 + 2x^2 - x + 1$

62. If [x] denotes the integral part of x and in $(0, \pi)$, we define

$$f(x) = \begin{cases} \left[\dfrac{2(\sin x - \sin^n x) + |\sin x - \sin^n x|}{2(\sin x - \sin^n x) - |\sin x - \sin^n x|}\right], & x \neq \dfrac{\pi}{2} \\ 3, & x = \dfrac{\pi}{2} \end{cases}$$

Then for $n > 1$,

(A) $f(x)$ is continuous but not differentiable at $x = \dfrac{\pi}{2}$

(B) both continuous and differentiable at $x = \dfrac{\pi}{2}$

(C) neither continuous nor differentiable at $x = \dfrac{\pi}{2}$

(D) $\lim\limits_{x \to \frac{\pi}{2}} f(x)$ exists but $\lim\limits_{x \to \frac{\pi}{2}} f(x) \neq f\left(\dfrac{\pi}{2}\right)$

63. Let f be a function defined on R by $f(x) = [x] + \sqrt{x - [x]}$, then

(A) f is not continuous at every $x \in I$ (B) f is not continuous at every $x \in R \sim I$

(C) f is a continuous function (D) none of these

Space for Rough work

64. The maximum value of area of rectangle *OABC*, where *O* is origin, *A* lies on positive *y*-axis and *B* lies on the curve $y = \frac{\ln x}{x^2}$, is

(A) e^{-1} (B) $e^{-1/2}$

(C) 1 (D) $e^{1/2}$

65. Let $f(x) = \int \frac{x^2 dx}{(1+x^2)(1+\sqrt{1+x^2})}$ and $f(0) = 0$. Then *f* (1) is

(A) $\log(1+\sqrt{2})$ (B) $\log(1+\sqrt{2}) - \frac{\pi}{4}$

(C) $\log(1+\sqrt{2}) + \frac{\pi}{4}$ (D) none of these

66. For $x > -1$, let $f(x) = \int_0^{\pi/4} \log_e(1 + x\tan z)dz$. Then value of $f\left(\frac{1}{2}\right) - f\left(\frac{1}{3}\right)$ equals

(A) $\frac{\pi}{4}\log_e\left(\frac{9}{8}\right)$ (B) $\frac{\pi}{8}\log_e\left(\frac{9}{8}\right)$

(C) $\frac{\pi}{9}\log_e\left(\frac{8}{9}\right)$ (D) $\frac{\pi}{8}\log_e\left(\frac{3}{2}\right)$

67. Area bounded by $y = \tan^{-1}x$, $y = \cot^{-1}x$ and *y*-axis is equal to

(A) $\ln\sqrt{2}$ sq. units (B) ln 4 sq. units

(C) ln 8 sq. units (D) ln 2 sq. units

68. Solution of $\sec^2 y\frac{dy}{dx} + x\tan y = x^3$ is

(A) $\tan y = x^2 + ce^{-x^2}$ (B) $\tan y = x^2 - 2 + ce^{-x^2}$

(C) $\tan y = x^2 - 2 + ce^{-x^2/2}$ (D) none of these

Space for Rough work

69. If $y = \dfrac{\cos 6x + 6\cos 4x + 15\cos 2x + 10}{\cos 5x + 5\cos 3x + 10\cos x}$, then $\dfrac{dy}{dx}$ is equal to

(A) $2\sin x + \cos x$ (B) $-2\sin x$

(C) $\cos 2x$ (D) $\sin 2x$

70. The tangents at z_1, z_2 on the circle $\left| z - z_0 \right| = a$, meet at z_3. Then $\left(\dfrac{z_3 - z_1}{z_0 - z_1}\right)\left(\dfrac{z_0 - z_2}{z_3 - z_2}\right)$ is equal to

(A) 0 (B) 1

(C) –1 (D) 2

71. The number of permutations of letters of word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears are

(A) 12246 (B) 169194

(C) 166680 (D) 181434

72. *A* is a 3 × 3 matrix with entries from the set {–1, 0, 1}. Then the probability that *A* is neither symmetric nor skew-symmetric is

(A) $\dfrac{3^9 - 3^6 - 3^3 + 1}{3^9}$ (B) $\dfrac{3^9 + 3^6 - 3^3 + 1}{3^9}$

(C) $\dfrac{3^9 - 3^6 + 3^3 + 1}{3^9}$ (D) none of these

73. Number of value(s) of *x* satisfying the equation $3x^2 - 2x^3 = \log_2\left(x^2 + 1\right) - \log_2 x$ is

(A) 1 (B) 2

(C) 3 (D) none of these

74. If $(1 + x + x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then the value of $\sum_{k=0}^{3r} (-1)^k a_k \, {}^nC_{3r-k}$ (where $3r \le n$) equals

(A) $(-1)^r . {}^nC_r$ (B) ${}^nC_{3r}$

(C) $(-1)^{3r} . {}^nC_{3r}$ (D) ${}^nC_r$

Space for Rough work

75. If $x \in R$ and $n \in I$, then the determinant $\Delta = \begin{vmatrix} \sin(n\pi) & \sin x - \cos x & \log\tan x \\ \cos x - \sin x & \cos\left[(2n+1)\frac{\pi}{2}\right] & \log\cot x \\ \log\cot x & \log\tan x & \tan(n\pi) \end{vmatrix}$ equals

(A) 0 (B) $\log\tan x - \log\cot x$

(C) $\tan\left(\frac{\pi}{4} - x\right)$ (D) none of these

76. If $A = \begin{bmatrix} 2 & 3 & 0 \\ 1 & \sin^2 x & -\cos^2 x \\ \cos^2 x & \cos^4 x & \sin^4 x \end{bmatrix}$ and $\text{A (Adj A)} = \text{K}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then the value of K is

(A) 0 (B) –1

(C) 1 (D) None of these

77. If *m* is the root of the equation $(1-ab)x^2 - (a^2+b^2)x - (1+ab) = 0$ and *m* harmonic means are inserted between *a* and *b*, then the difference between the last and the first of the means equals

(A) $ab(a+b)$ (B) $a(b-a)$

(C) $ab(b-a)$ (D) $b-a$

78. Let S_n denote the sum to *n* terms of an arithmetic progression. If the common difference is equal to $S_n - kS_{n-1} + S_{n-2}$, then *k* =

(A) 1 (B) 2

(C) 3 (D) none of these

79. If the equal sides *AB* and *AC* along positive *x* and *y* axes (each equal to *a*) of a right angled isosceles triangle *ABC* be produced to *P* and *Q* respectively so that *B*P. *CQ* = AB^2, then the line *PQ* always passes through a fixed point

(A) (*a*, 0) (B) (0, *a*)

(C) (*a, a*) (D) none of these

Space for Rough work

80. The equation of the circle, which is touched by the line $y = x$, has its centre on the positive direction of the x-axis and cut off a chord of length 2 units on the line $\sqrt{3}\ y = x$, is

(A) $x^2 + y^2 - 4x + 2 = 0$ (B) $x^2 + y^2 - 4x + 1 = 0$

(C) $x^2 + y^2 - 8x + 8 = 0$ (D) $x^2 + y^2 - 4y + 2 = 0$

81. If the tangents at two points (1, 2) and (3, 6) on a parabola intersect at the point (1, 6), then equation of directrix is

(A) $2x + y - 8 = 0$ (B) $x - 2y + 11 = 0$

(C) $x + 2y - 13 = 0$ (D) $2x - y + 4 = 0$

82. If tangent and normal at point P (in first quadrant) to ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ intersect major axis at T and N respectively in such a way that ratio of area of ΔPTN and $\Delta PSS'$ is $\frac{91}{60}$, then area of $\Delta PSS'$ is (S and S' are foci)

(A) $6\sqrt{3}$ sq. units (B) $12\sqrt{3}$ sq. units

(C) $4\sqrt{3}$ sq. units (D) $3\sqrt{3}$ sq. units

83. If a hyperbola whose foci are $S \equiv (2, 4)$ and $S' \equiv (8, -2)$ touches x-axis, then equation of hyperbola is

(A) $\frac{(x+y-6)^2}{10} - \frac{(x-y-4)^2}{8} = 1$ (B) $\frac{(x-y-4)^2}{10} - \frac{(x+y-6)^2}{8} = 1$

(C) $\frac{(x-y-4)^2}{20} - \frac{(x+y-6)^2}{16} = 1$ (D) $\frac{(x+y-6)^2}{20} - \frac{(x-y-4)^2}{16} = 1$

Space for Rough work

84. If $\sin A = x,\ \cos B = y,\ A + B + C = 0,$ then $x^2 + 2xy\sin C + y^2$ is equal to

(A) $\sin^2 C$ (B) $\cos^2 C$

(C) $1 + \sin^2 C$ (D) $1 + \cos^2 C$

85. The values of $x \in \left(0, \frac{\pi}{2}\right)$ satisfying the equation $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ are

(A) $\frac{\pi}{36}, \frac{\pi}{12}$ (B) $\frac{\pi}{12}, \frac{11\pi}{36}$

(C) $\frac{12\pi}{36}, \frac{11\pi}{36}$ (D) $\frac{11\pi}{36}, \frac{\pi}{36}$

86. If in $\Delta ABC,\ \sin B\cos C = \frac{\sqrt{3}-1}{\sqrt{3}}$ and $\cos B\sin C = \frac{1}{\sqrt{3}}$, then the triangle is

(A) isosceles (B) equilateral

(C) right angled (D) right angled isosceles

87. If for $x < -1, \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) + \sin^{-1}\left(\frac{2x}{1+x^2}\right) - \tan^{-1}\left(\frac{2x}{x^2-1}\right) = \frac{\pi}{3},$ then x is equal to

(A) $-\sqrt{3}$ (B) $-\sqrt{2}$

(C) –2 (D) none of these

88. The vector equation of the plane through the point $2\hat{i} - \hat{j} - 4\hat{k}$ and parallel to the plane $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) - 7 = 0$ is

(A) $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) = 0$ (B) $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) = 32$

(C) $\vec{r}\,.(4\hat{i} - 12\hat{j} - 3\hat{k}) = 12$ (D) none of these

89. If statement $(p \to q) \to (q \to r)$ is false, then truth values of statement p,q,r respectively, can be-

(A) FTF (B) TTT

(C) FFF (D) FTT

90. Standard Deviation of a data is 7, when each observation is decreased by 7 then standard deviation of new data is

(A) 0 (B) 14

(C) 7 (D) 49

Space for Rough work